

Milestone 1 Report
ASCEND: Adaptive Self-supervised Continual Expert Network
for Domain-Incremental Wi-Fi Sensing
ECE 310 - Wireless Communication, Monsoon Semester 2026

Student	Dhvanit Mahesuriya (AU2440039)
Programme	B.Tech Computer Science Engineering
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1. Problem Statement

Wi-Fi sensing uses Channel State Information (CSI) to detect human activities. It does not need a camera or a wearable device. Its performance can change when the environment changes.

The main problem is domain-incremental Wi-Fi Human Activity Recognition (HAR). The model should learn new environments over time. It should also keep what it learned before. A key problem is catastrophic forgetting. The model may forget old domains after learning a new domain.

Labelled CSI data is also limited. Collecting labels for every new environment takes time. It also needs manual work. Storing old samples can create privacy and storage issues.

This work studies whether a Wi-Fi model can learn new environments with fewer labels. It also studies how to reduce forgetting. Self-supervised learning and a mixture-of-experts model are used for this purpose.

Motivation

Wi-Fi sensing can work without cameras. This makes it useful for activity recognition. A practical model should not need full retraining whenever the environment changes.

Continual learning can help the model learn new domains. Self-supervised learning can reduce the need for labels. A mixture-of-experts (MoE) model can learn domain-specific features.

Significance

The planned framework combines self-supervised learning and continual learning for Wi-Fi HAR. It uses different experts for different domains. The main goal is to learn new domains without losing old knowledge.

2. Review of Existing Work

The initial literature study focused on recent work covering continual Wi-Fi sensing, cross-domain HAR, self-supervised Wi-Fi sensing, and mixture-of-experts approaches. The main papers studied so far are summarized below.

Work	Problem / Focus	Main Idea	Relevance to ASCEND
ConSense (2025)	Continual human activity sensing with Wi-Fi	Continual sensing with growing and picking.	Shows the need for continual learning.
Wi-DMAR (2026)	Cross-domain human activity recognition	Conditional diffusion for cross-domain HAR.	Relevant to domain shift.
CIG-MAE (2026)	Self-supervised Wi-Fi sensing	Cross-modal masked autoencoder.	Supports self-supervised learning with fewer labels.
Scene-adaptive continual learning with MoE (2026)	Continual CSI-based HAR under changing scenes	Scene-adaptive continual learning with MoE.	Useful reference for continual MoE learning..

Key Observations from Literature

- Catastrophic forgetting is a central issue in Wi-Fi sensing.
- Label availability remains a bottleneck for Wi-Fi sensing systems.

- Keeping exemplars from old domains can introduce privacy and storage concerns.
- Cross-domain Wi-Fi HAR has received attention, but continual domain adaptation with limited labelled data remains an open direction.
- Self-supervised learning provides a possible way to learn useful representations without requiring labels for every sample.
- Mixture-of-experts architectures provide a possible mechanism for allocating different model capacity to different domains.

3. Current SOTA Position and Research Direction

The work combines four areas: Wi-Fi CSI sensing, domain adaptation, continual learning, and self-supervised learning. Previous works have studied these areas separately or in different combinations. Among the existing works, SAMoE-C is the closest SOTA work to our problem, as it considers continual learning for Wi-Fi sensing across domains but follows supervised learning. However, it does not combine all four areas in the way proposed in this work. Therefore, this work aims to bring these areas together to improve the reliability of Wi-Fi sensing under changing domains.

The tentative framework is named ASCEND, standing for Adaptive Self-supervised Continual Expert Network for Domain-Incremental Wi-Fi Sensing. The central idea is to use self-supervised learning to obtain a useful representation, then adapt the model to incoming domains using expert-based continual learning while attempting to preserve earlier knowledge.

Initial Conceptual Components

- Self-supervised learning: learn representations from CSI data without requiring labels for every sample.
- Domain-incremental learning: process new sensing environments sequentially rather than assuming that all domains are available at once.
- Continual expert learning: use expert capacity to model domain-specific characteristics while reducing interference between domains.
- Forgetting control: evaluate whether adaptation to later domains causes a significant loss on earlier domains.

4. Progress Completed

Area	Status	Progress
Domain understanding	Completed	Studied Wi-Fi sensing using CSI for human activity recognition and identified domain shift as a major practical issue.
Problem formulation	Completed	Defined the target problem around domain-incremental Wi-Fi HAR, limited labels, and catastrophic forgetting.
Literature review	Completed / ongoing refinement	Reviewed recent works including ConSense, Wi-DMAR, CIG-MAE, and a scene-adaptive continual MoE approach.
Research gap identification	Completed	Identified the need to study continual domain adaptation in Wi-Fi HAR while reducing dependence on labelled data and stored exemplars.
MoE understanding	Completed	Studied the basic mixture-of-experts concept and its relevance to domain-specific learning.
Baseline/KPI planning	Started	Identified the need to reproduce relevant baselines and determine suitable continual-learning and HAR metrics.
ASCEND development	Started	Initial coding and experimental preparation have begun; the framework and experiments are being developed further.

Paper direction	Defined	Tentative title, research direction, and publication roadmap have been established.
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At this stage, the problem and research gap are defined. The main papers have been studied. Baseline planning is also done. Initial implementation has started. The experiments are still in progress.

5. Planned Experimental Direction

- Use suitable Wi-Fi CSI HAR datasets and define an ordered sequence of domains/environments for continual learning.
- Reproduce relevant baseline methods before making comparisons with ASCEND.
- Track both current-domain performance and retention of previously learned domains.
- Study the effect of limited labelled data and investigate whether self-supervised pretraining reduces the label requirement.
- Evaluate catastrophic forgetting using appropriate retention/forgetting measures in addition to standard classification accuracy.
- Study the effect of expert allocation and domain adaptation components through ablation experiments.
- Perform hyperparameter sweeps only after the baseline implementation and core ASCEND pipeline are stable.

6. Timeline and Next Milestones

Phase	Planned Work	Status
Phase 1 (Weeks 1-3)	Understanding domain; finding problem formulation; critically studying recent literature.	Completed
Phase 2 (Weeks 4-6)	Reproducing baselines; understanding possible KPIs; starting ASCEND coding.	In progress
Phase 3 (Weeks 7-9)	Running experiments; generating relevant KPIs; hyperparameter sweeping.	Upcoming
Phase 4 (Week 10)	Final paper preparation and final presentation.	Planned

Immediate Next Steps

1. Complete and verify baseline reproductions.
2. Finalize the evaluation metrics and continual domain split.
3. Complete the first stable ASCEND implementation.
4. Run initial experiments and check whether the expected forgetting/adaptation behavior is observed.
5. Generate KPI plots and compare ASCEND with the selected baselines.
6. Use ablation studies to determine which components contribute to the final performance.

7. Expected Evaluation

The evaluation will check recognition performance and forgetting. Accuracy will be measured for each domain. The model will also be checked on old domains after learning new ones.

Evaluation Aspect	Purpose	Stage
HAR performance	Measure activity recognition quality on each domain.	Baseline and final evaluation
Cross-domain generalization	Measure robustness when the sensing domain order changes.	Main experiments
Catastrophic forgetting	Measure performance degradation on earlier domains after learning new domains.	Main continual-learning evaluation

Label efficiency	Study performance under limited labelled data.	Ablation / sensitivity study
Ablation	Measure the contribution of self-supervision, experts, and continual-learning components.	Final experiments

8. Conclusion

The first milestone established the research direction for ASCEND. The work has moved from domain understanding to a clear problem formulation, literature review, research-gap identification, and initial framework planning. The current focus is shifting toward baseline reproduction and implementation. The next milestone will concentrate on obtaining reliable experimental results and KPIs, followed by systematic comparison and ablation studies.

References

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